Backing up the SSD system

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During development, users may need to back up the system so that later development does not affect the current system environment.

```
The image file demonstrated in this tutorial is not the actual factory image; it is used for tutorial demonstration only
```

1. Hardware connections

Users need to prepare an SSD enclosure in advance, install the SSD into the enclosure and connect it to the computer or the virtual machine: the computer and the virtual machine must run an Ubuntu system.



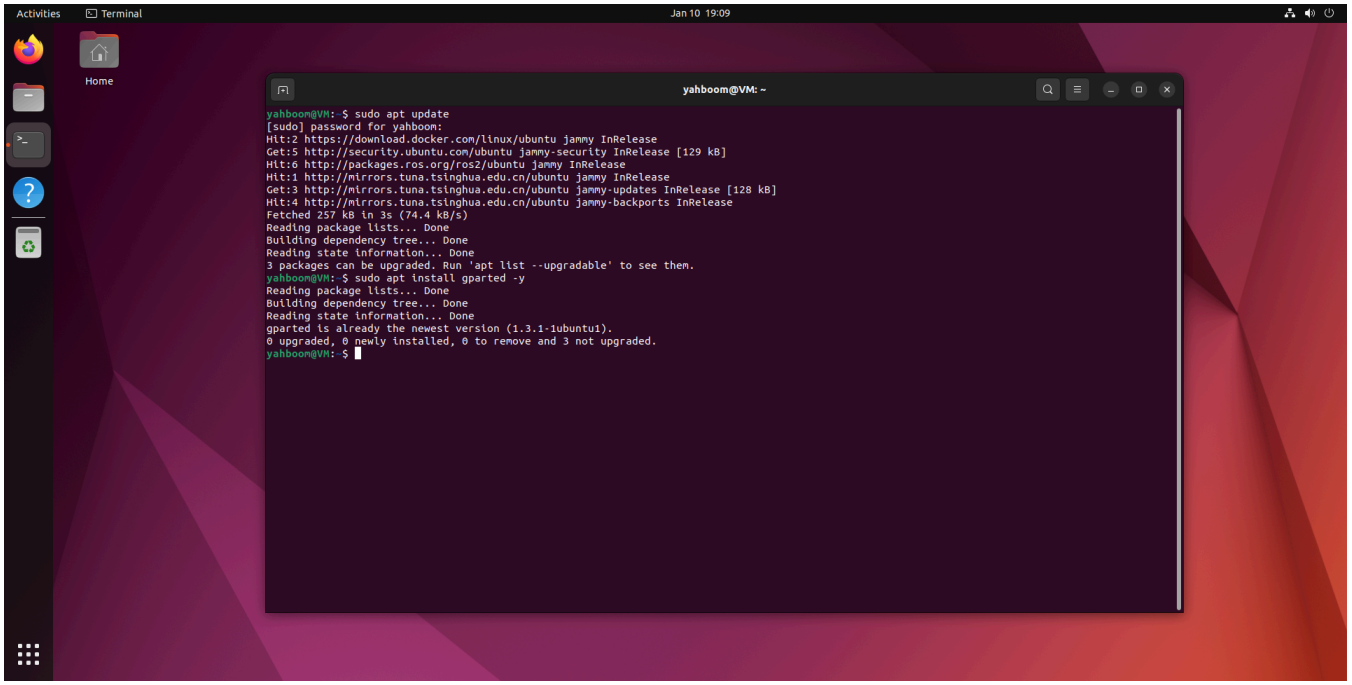


2. Compressing the SSD

Since the SSD bundled with the Jetson Orin series mainboards has a relatively large capacity, we need to shrink it to a suitable size for the system backup, which saves time on backing up and flashing the system.

2.1. Installing Gparted

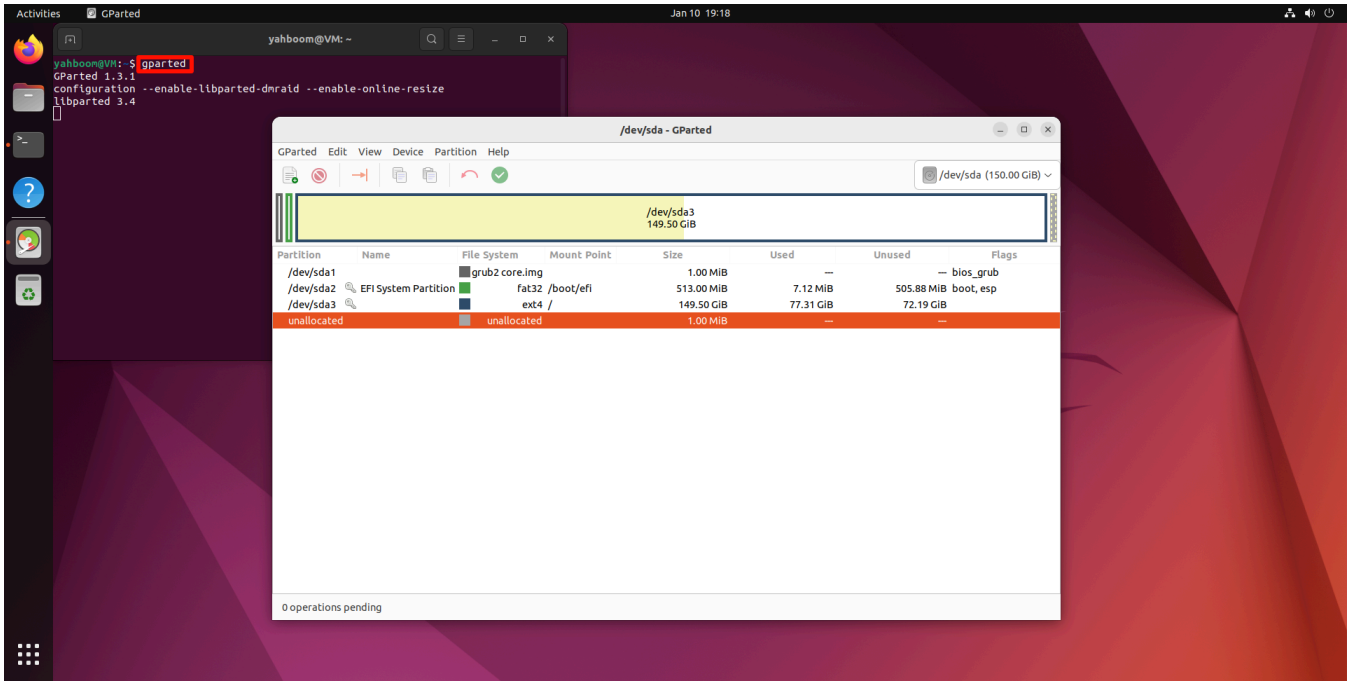
```
sudo apt update
sudo apt install gparted -y
```



2.2. Using GParted

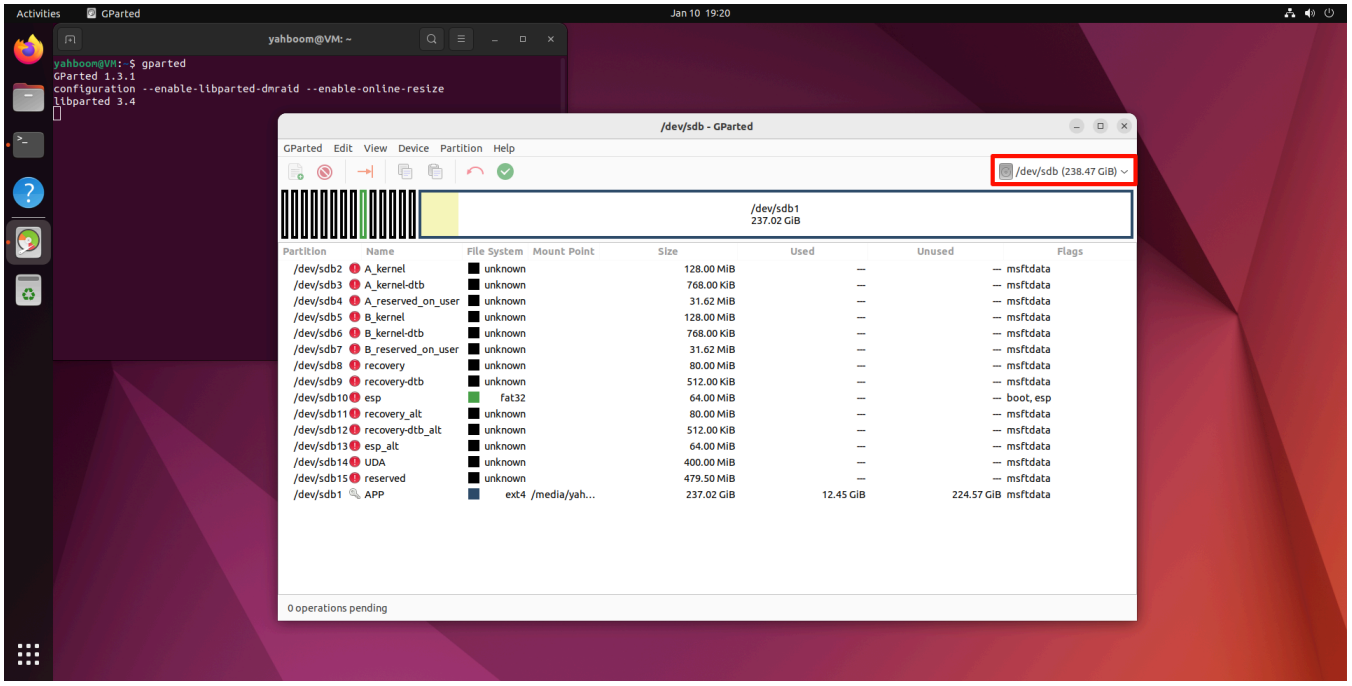
Find the `GParted` application icon in the system application menu and open it, or launch it by entering the following command in a terminal:

```
gparted
```



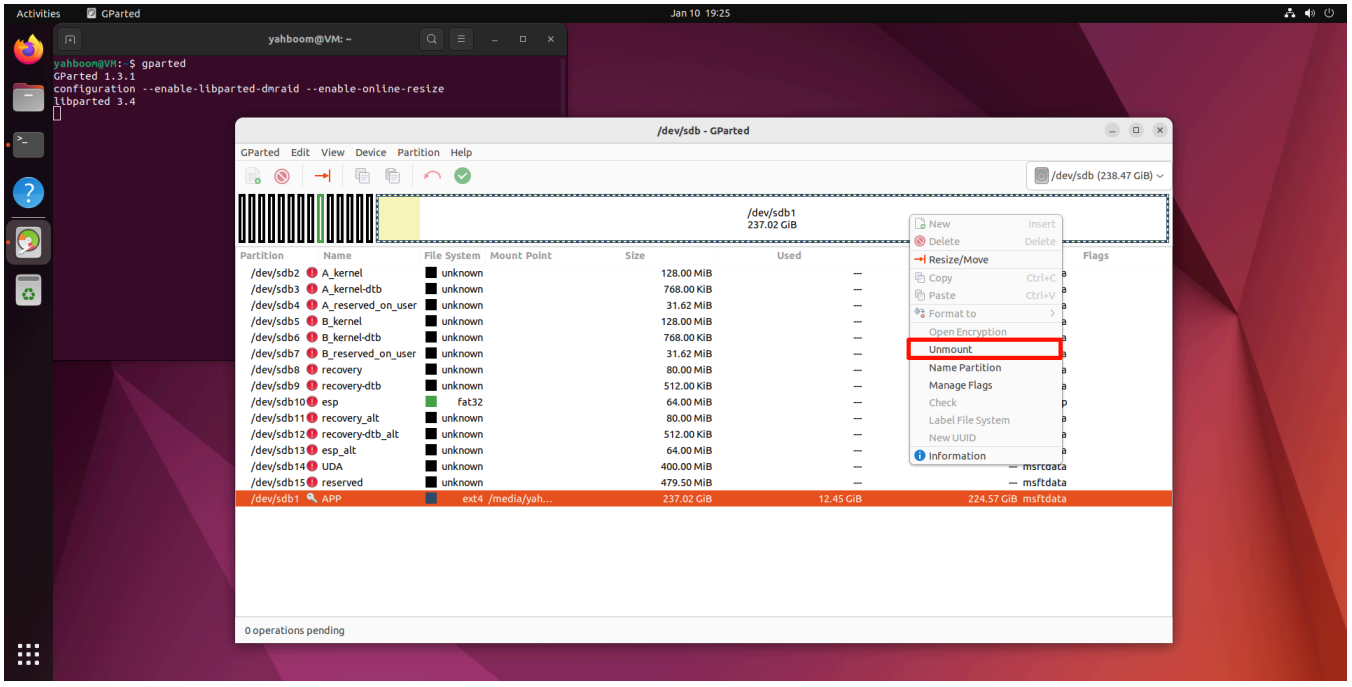
2.2.1. Selecting the SSD

Select the newly added disk icon: you can confirm once more, by the disk capacity, whether it is the SSD you mounted



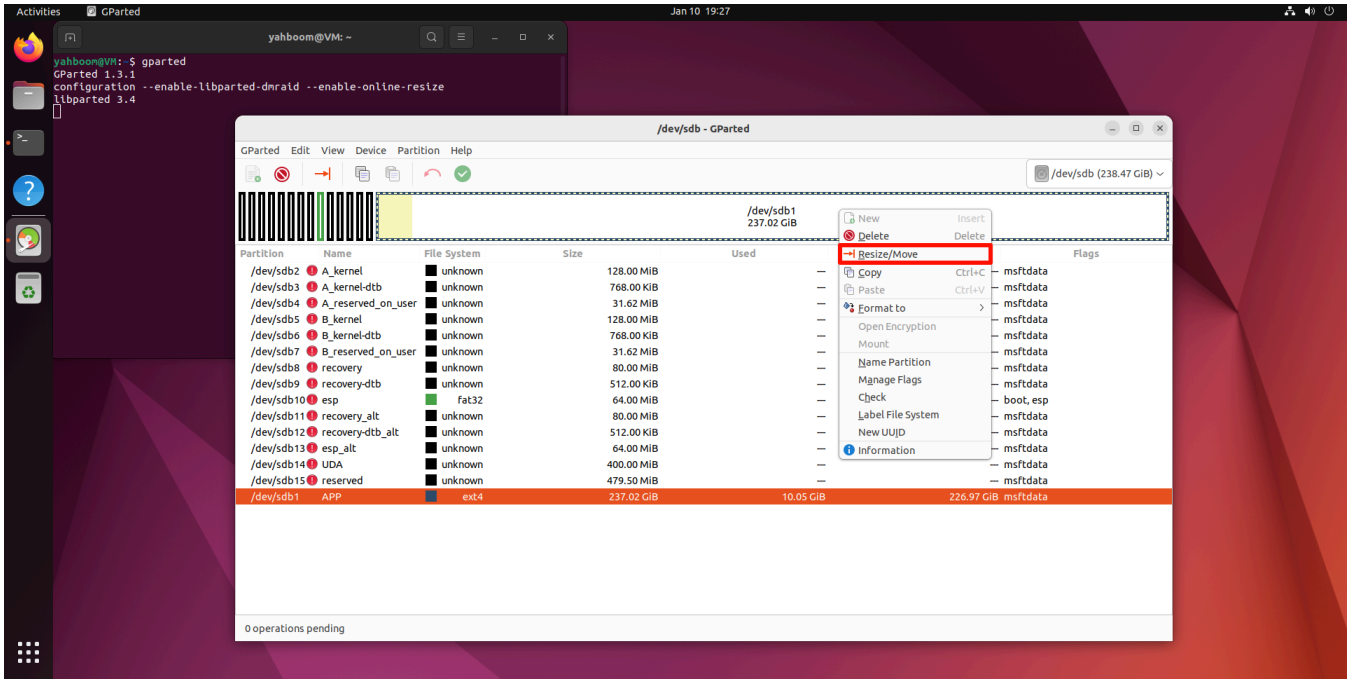
2.2.2. Unmounting the partition

Before operating on the disk, you must first unmount it: select the `APP` partition in the disk (the largest partition) and click `Unmount` to unmount the partition

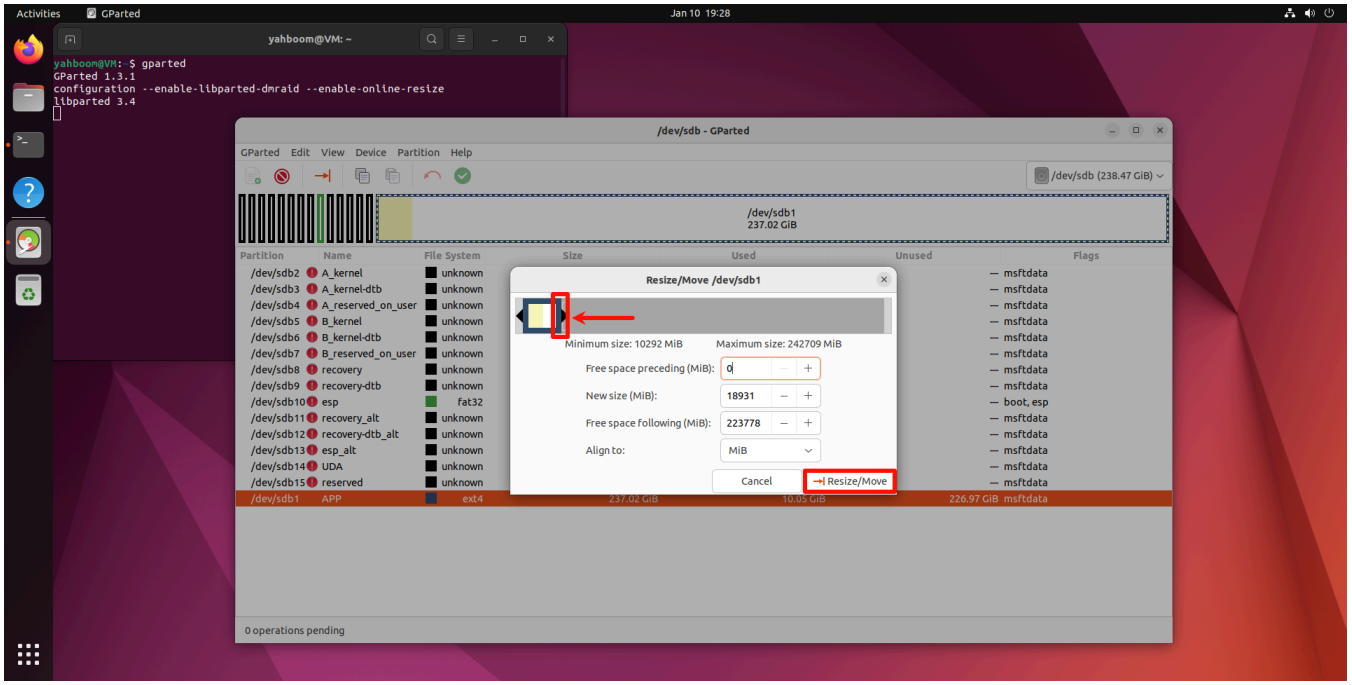


2.2.3. Shrinking the disk

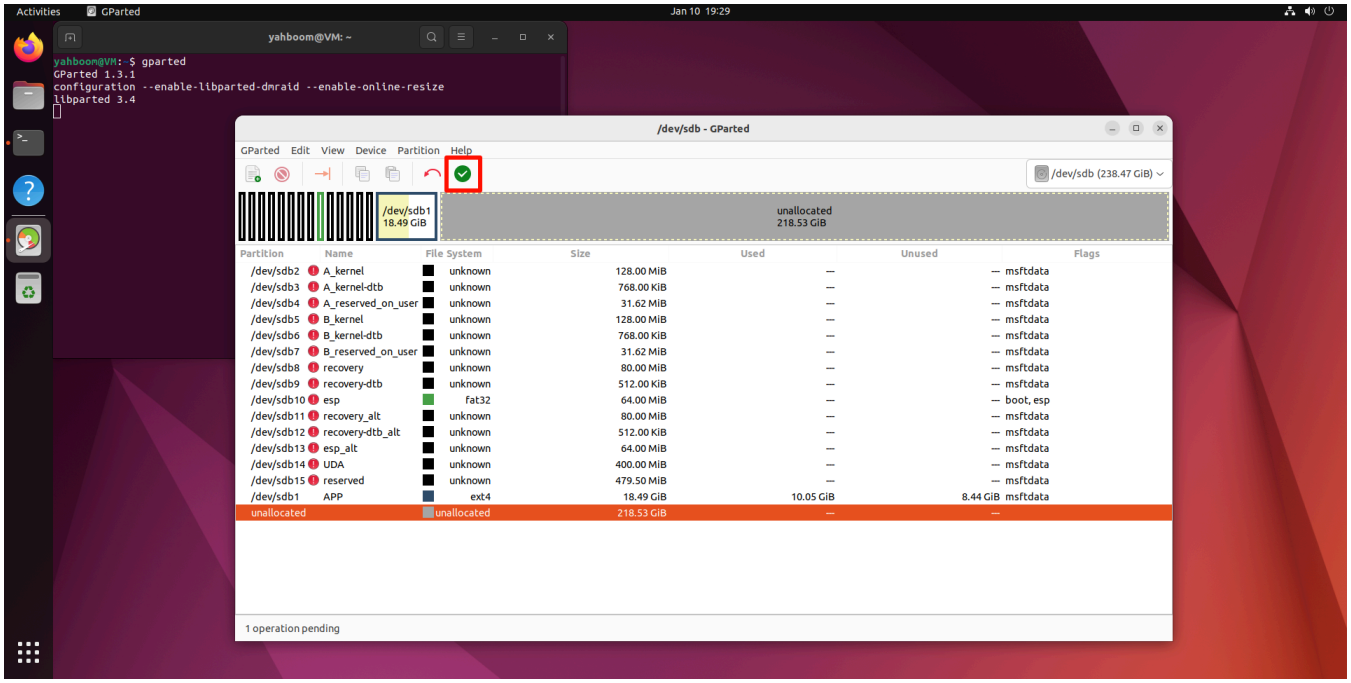
Right-click the unmounted disk partition with the mouse and resize the space of the partition that was unmounted earlier:

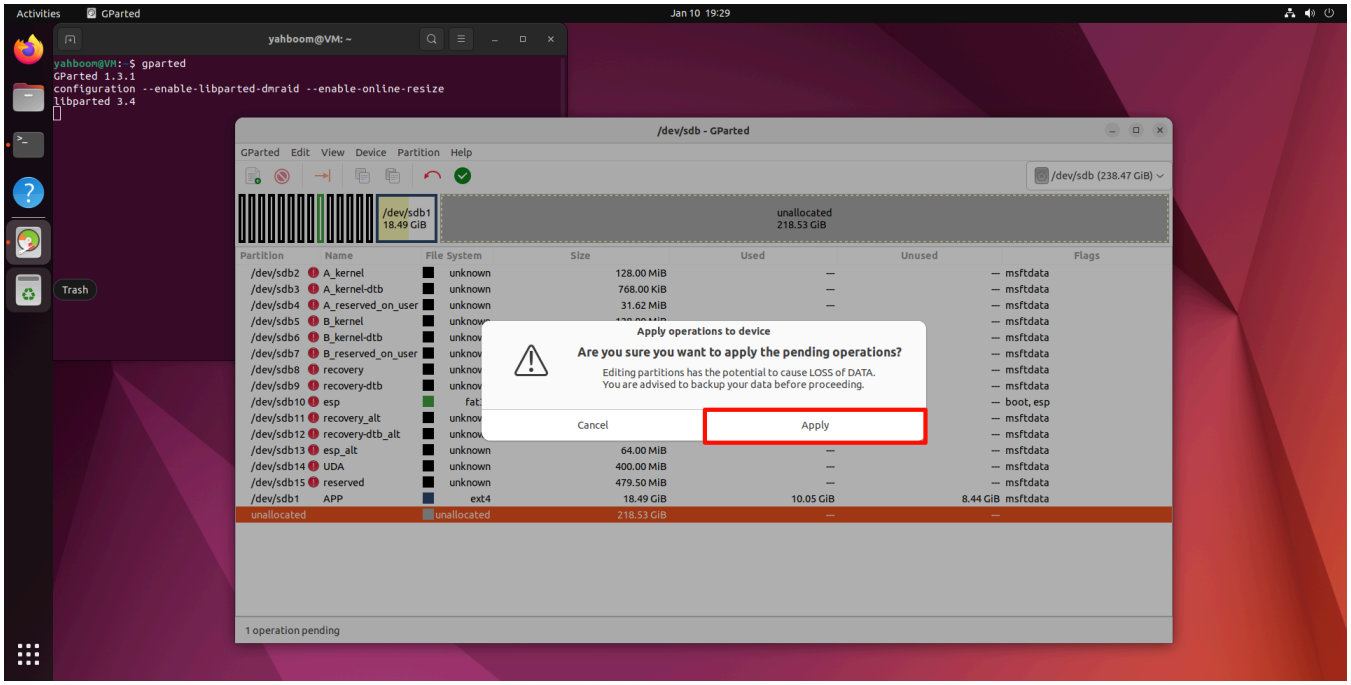


You can adjust the partition size with the slider: yellow is the space used by the partition and white is the unused space; it is recommended to leave about 5-10G of unused space in the partition to avoid the system failing to boot

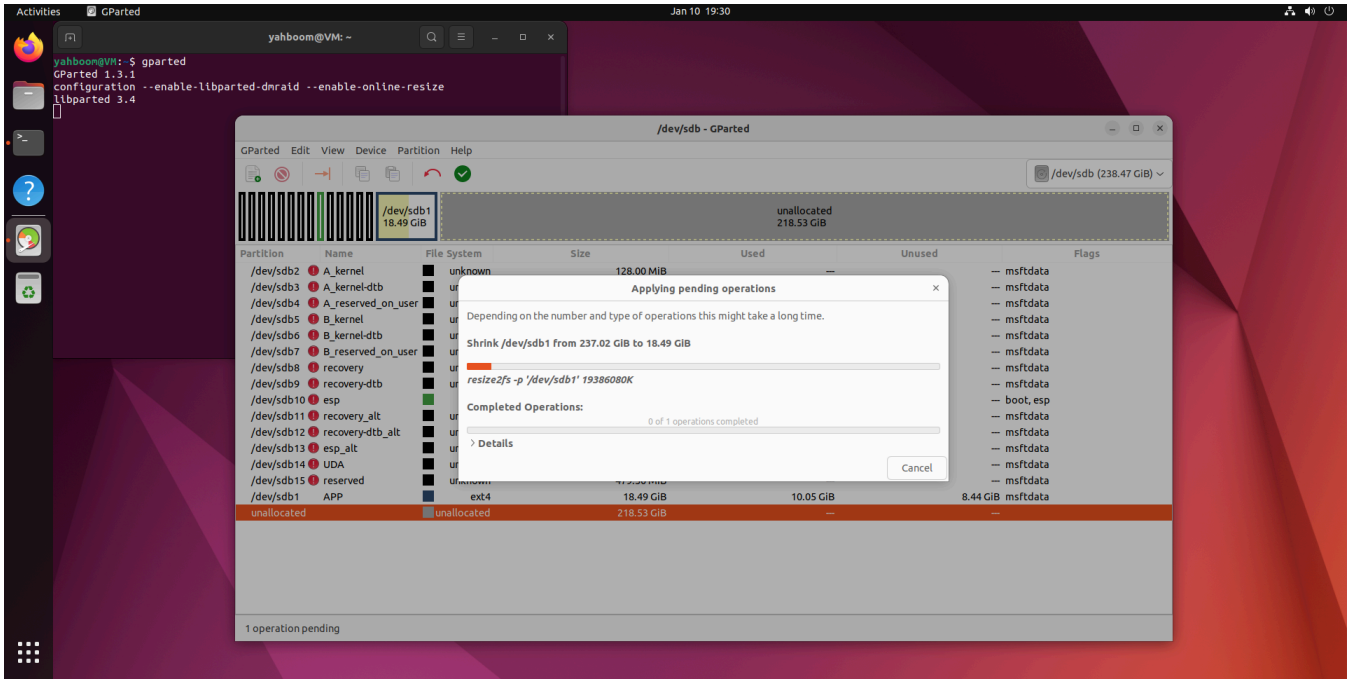


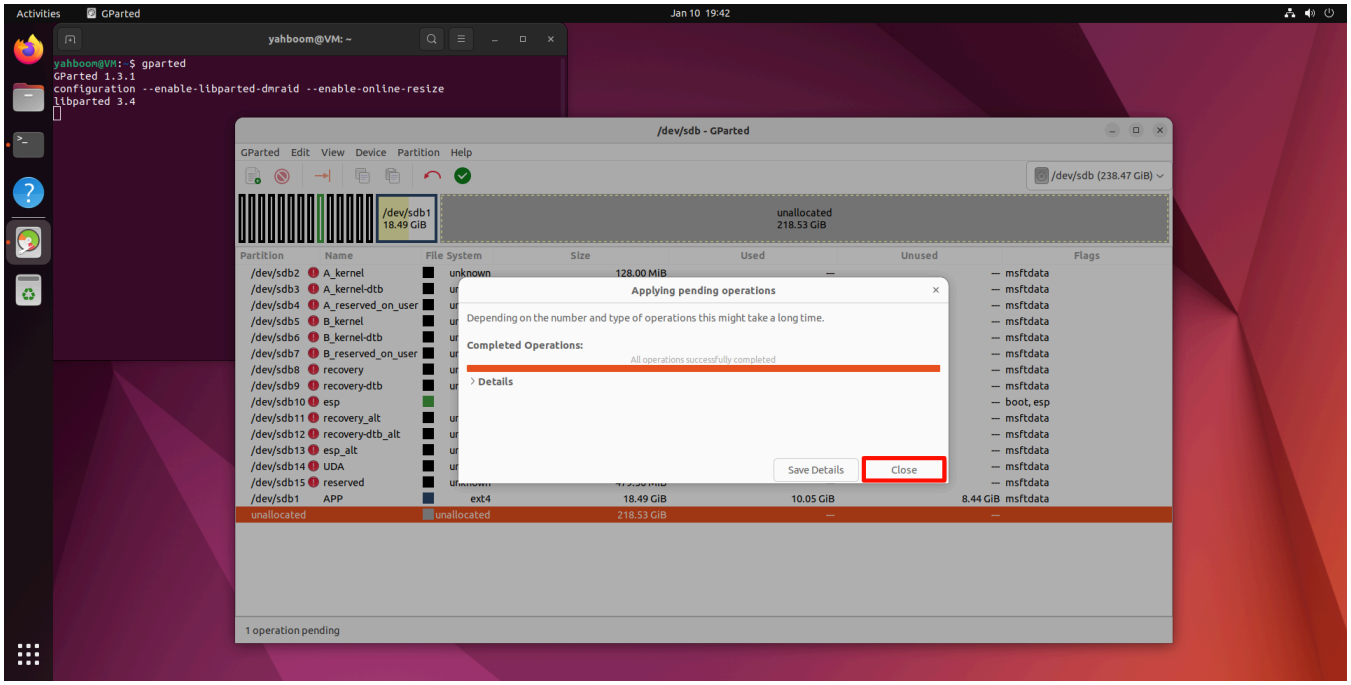
Confirm the disk operation:





Wait for the operation to complete:





After completing the above operations, close GParted!

3. Backing up the SSD

3.1. Viewing disk information

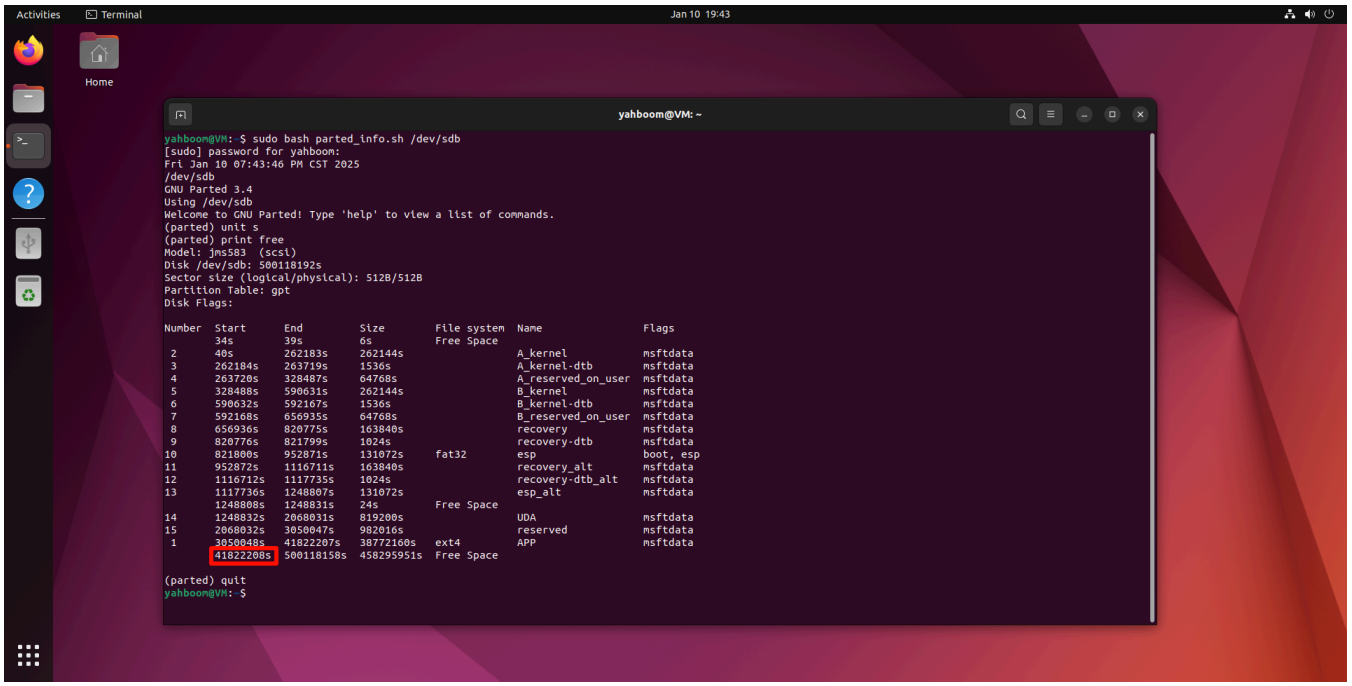
Open a terminal and use the script to view the current disk information: the device name must correspond to that of the SSD you are backing up

```
sudo bash parted_info.sh /dev/sdb
```

> **Content of the parted_info.sh script**

```
#!/bin/bash
date
echo $1
sudo parted $1 <<EOF
unit s
print free
quit
EOF
```

Record the value shown in the image: 41822208s



3.2. Starting the disk backup

Use the `dd` command to back up the SSD into an `img` file; enter the following in a terminal:

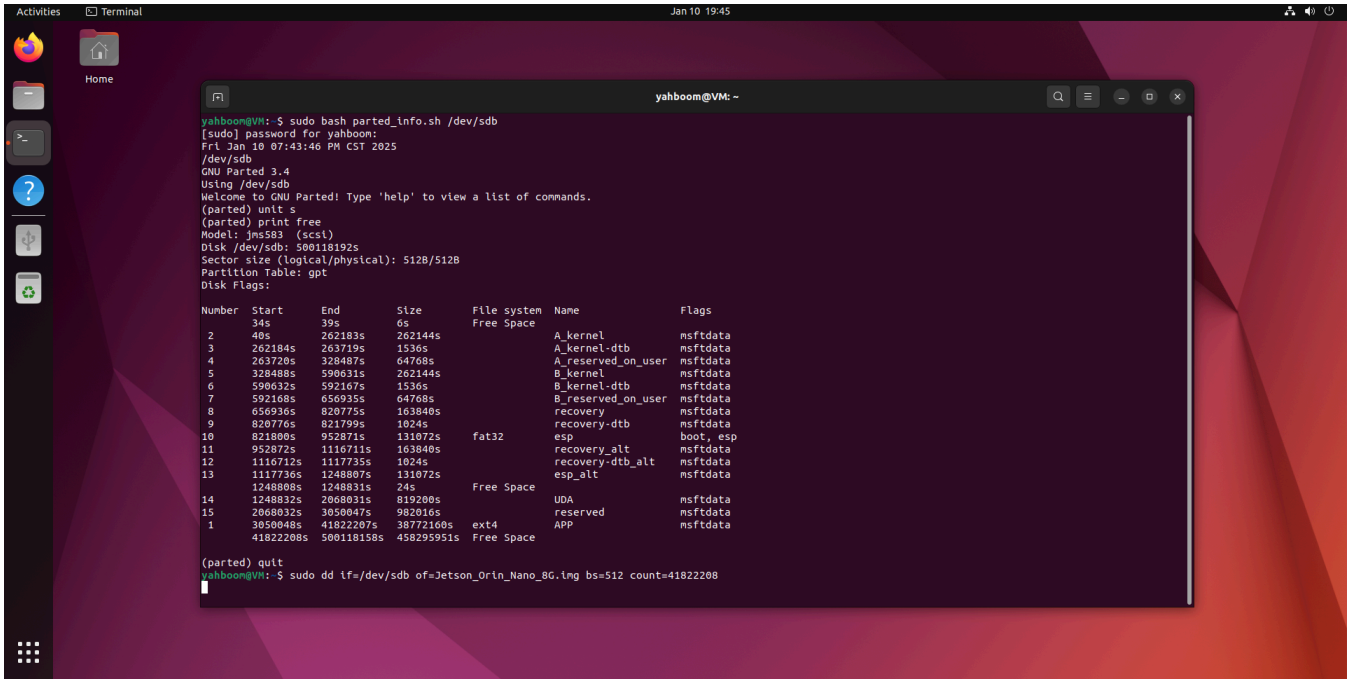
```
sudo dd if=/dev/sdb of=Jetson_Orin_Nano_8G.img bs=512 count=41822208
```

`/dev/sdb`: the SSD device name

`Jetson_Orin_Nano_8G.img`: the image file name

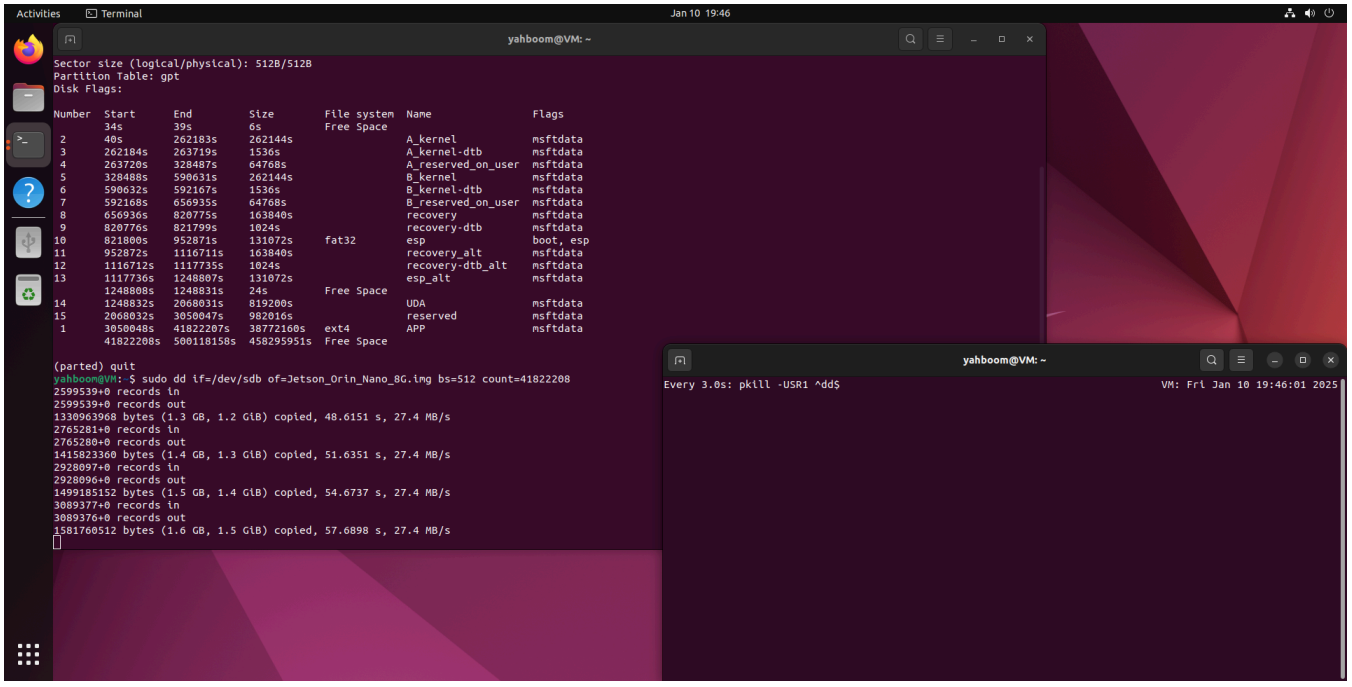
`bs=512`: set the block size to 512 bytes

`56393728`: the value queried by the script

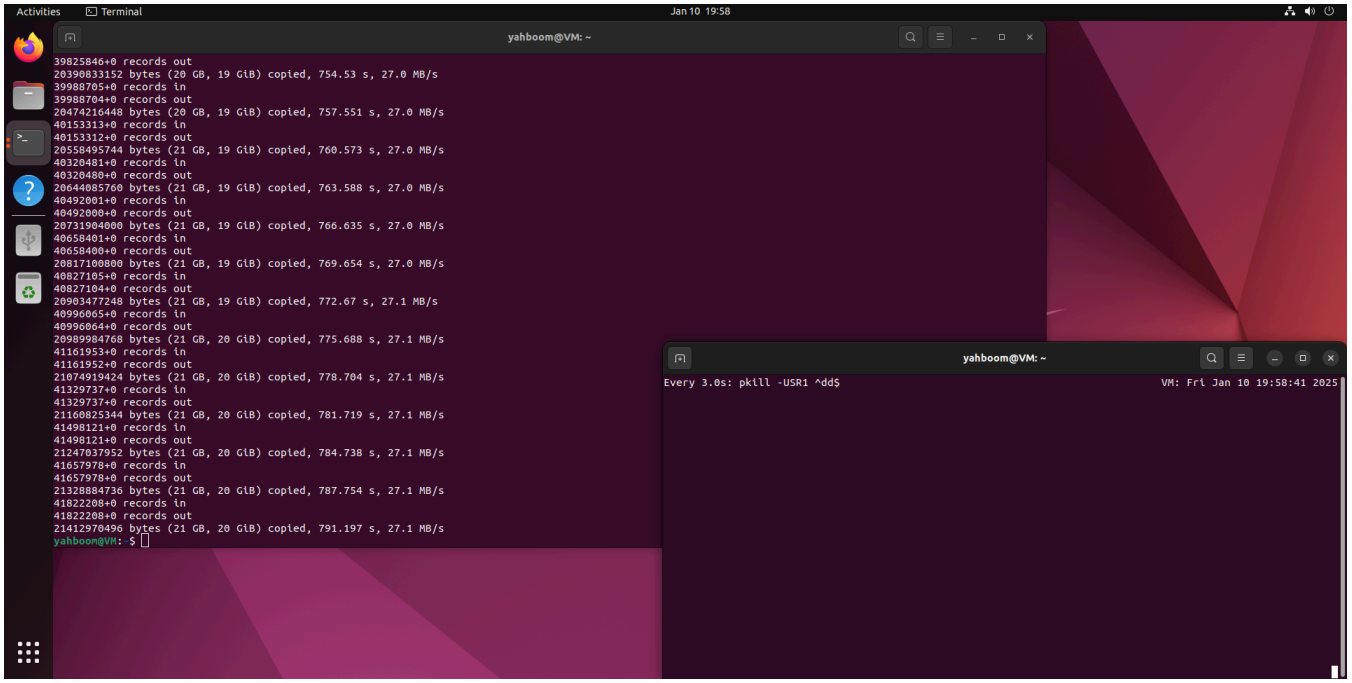


To view the `dd` process information, open another terminal and enter the following command:

```
sudo watch -n 3 pkill -USR1 ^dd$
```



Wait for the backup to complete:



After the system backup is complete, move the backup file (Jetson_Orin_Nano_8G.img) to a Windows system for use.

